Design of high gain antenna array for mm-wave 5G Communications

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submitted by

Name : Devashish Tushar (2022UEC1487)

Name : Aditya Kulshreshtha (2022UEC1517)

Name : Puneet Kumar (2022UEC1673)

Name : Ashish Kumar (2022UEC1698)

Name : Shivansh Gupta (2022UEC1637)

Under the Supervision of

Dr. Reena Kumari

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

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CERTIFICATE

This is to certify that the VII Semester Minor Project (ECD-498) report entitled "**Design of high gain antenna array for mm-wave 5G Communications**" submitted by Name : Devashish Tushar (Roll Number : 2022UEC1487); Name : Aditya Kulshreshtha (Roll Number : 2022UEC1517); Name : Puneet Kumar (Roll Number :2022UEC1673); Name : Ashish Kumar(Roll Number : 2022UEC1698); Name : Shivansh Gupta (Roll Number : 2022UEC1637) is a genuine project work carried out under my supervision.

(Signature)

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Supervisor : Dr. Reena Kumari

Dept. of Electronics and Communication Engineering

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Name : Devashish Tushar (Roll Number : 2022UEC1487)

Name : Aditya Kulshreshtha (Roll Number : 2022UEC1517)

Name :Puneet Kumar (Roll Number :2022UEC1673)

Name : Ashish Kumar(Roll Number : 2022UEC1698)

Name : Shivansh Gupta (Roll Number : 2022UEC1637)

ABSTRACT

The evolution of 5G communication networks, particularly in the millimeter-wave spectrum, requires antenna systems that offer compact size, high gain, wide bandwidth, and strong isolation to support multi-gigabit data rates. Addressing these demands, this work presents the design and development of a compact array antenna system capable of operating efficiently at both 28 GHz and 38 GHz. The proposed structure is realized on a Rogers RT/Duroid 5880 substrate and incorporates a two-element array configuration excited through a corporate-feed network, enabling uniform signal distribution, improved impedance control, and enhanced phase stability. To further optimize performance, a metallic strip is strategically positioned at the patch edge while a circular slot is introduced in the ground plane, collectively contributing to improved impedance matching, increased bandwidth, and superior radiation characteristics in both frequency bands.

Comprehensive electromagnetic simulations followed by experimental validation confirm dual-band operation spanning 27.1–28.7 GHz and 36.2–42 GHz, along with consistently low mutual coupling—below −20 dB in all scenarios and even lower in the four-port configuration. The array antenna structure demonstrates significant performance advantages, achieving maximum realized gains of 10.4 dBi at 28 GHz and 10.1 dBi at 38 GHz, while exhibiting directional radiation at the lower band and semi-omnidirectional behavior at the upper band. Furthermore, critical diversity parameters including Envelope Correlation Coefficient (ECC), Channel Capacity Loss (CCL), Total Active Reflection Coefficient (TARC), and Mean Effective Gain (MEG) remain within standard acceptable thresholds, ensuring high reliability in multipath environments. Owing to its compact footprint, strong isolation, and excellent dual-band radiation performance, the proposed antenna system stands out as a promising candidate for next-generation 5G mm-wave devices and high-capacity wireless communication platforms.

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Chapter 1

Introduction

1.1 Background and Motivation

The rapid evolution of wireless communication systems over the past decade has been primarily driven by the increasing demand for high-speed data services, real-time connectivity, enhanced mobility, and seamless support for billions of interconnected devices. While fourth-generation (4G LTE) networks significantly enhanced data capacity and user experience, the exponential rise of advanced applications such as high-definition multimedia streaming, augmented reality (AR), virtual reality (VR), and massive Internet of Things (IoT) networks has created a need for more advanced communication technologies. Fifth-generation (5G) networks have emerged to fulfill this requirement by offering multi-gigabit data rates, extremely low latency, and robust connectivity across heterogeneous devices and platforms.

To achieve these performance levels, access to wider bandwidths is essential, yet the sub-6 GHz spectrum is already congested and insufficient for supporting next-generation demands. This limitation has shifted global research and development efforts toward the millimeter-wave (mmWave) frequency band, spanning 24–100 GHz. Within this range, the

28 GHz and 38 GHz bands have gained widespread acceptance as primary candidates for 5G New Radio (NR), owing to their favorable propagation characteristics and greater spectrum availability. However, despite offering wide bandwidths, mmWave frequencies suffer from high free-space path loss, limited penetration ability, sensitivity to environmental blockages, and atmospheric absorption. These challenges directly influence antenna design, making the development of efficient mmWave antennas crucial for stable and high-performance communication.

Microstrip patch antennas have emerged as a practical solution for mmWave applications due to their low profile, lightweight structure, ease of fabrication, and excellent compatibility with RF front-end circuits. Advancements in low-loss substrates such as Rogers RT/Duroid 5880 have further enabled microstrip antennas to operate effectively at mmWave frequencies. However, designing dual-band microstrip antennas for 28 GHz and 38 GHz operation is highly challenging due to the very small wavelengths associated with mmWave signals. Achieving dual-band functionality requires intelligent design techniques such as patch modifications, slot loading, defected ground structures (DGS), and optimized feeding mechanisms.

While array technology significantly strengthens mmWave communication by enabling beamforming and spatial diversity, closely spaced antenna elements in compact devices introduce strong mutual coupling. This coupling degrades isolation, increases correlation, and reduces overall system performance. Therefore, the development of compact, high-performance, dual-band mmWave array antennas with low mutual coupling and stable radiation characteristics has become a critical research area. This project aims to address these requirements by developing an optimized array antenna design capable of supporting efficient 5G mmWave communication at both 28 GHz and 38 GHz.

1.2 5G mmWave and Dual-Band Requirements

The shift from sub-6 GHz communication to mmWave frequencies represents one of the most significant advancements in 5G technology. Conventional frequency bands are heavily utilized, offering limited bandwidth and insufficient capacity for emerging high-speed applications. In contrast, mmWave bands provide the extremely wide bandwidth needed to support multi-gigabit data transmission, dense user environments, and next-generation wireless services. Among these, the 28 GHz band, designated under the n257 band of 5G NR, is extensively used for fixed wireless access and small-cell deployments due to its comparatively better penetration capabilities and lower atmospheric absorption. The 38 GHz band, corresponding to the 3GPP n261 band, is particularly suitable for ultra-dense urban areas where high user capacity and extremely wide bandwidth are required.

A dual-band antenna operating at both 28 GHz and 38 GHz offers several advantages including multi-region compatibility, flexible carrier aggregation, better spectrum utilization, reduced hardware redundancy, and seamless switching between frequency bands during user mobility. However, designing such antennas is inherently challenging due to the extremely small wavelength—approximately 7 to 10 mm—at these frequencies. Small wavelengths lead to higher radiation losses, narrowband operation, increased sensitivity to physical geometry, and tighter fabrication tolerances. As a result, dual-band antenna designs require innovative techniques such as multi-resonant patch structures, carefully engineered ground-plane modifications, slot-based tuning, and high-precision feeding networks. Stability in radiation patterns and adequate gain at both resonant frequencies are also critical for ensuring reliable performance.

1.3 Antenna Design Challenges

Technology enhances wireless communication systems by using multiple antennas at the transmitter and receiver to improve throughput, spectral efficiency, and reliability. At mmWave frequencies,the array antenna becomes particularly important because highly directional beams are essential for overcoming severe free-space attenuation. However, integrating multiple antenna elements into compact mmWave devices introduces several design challenges.

The most significant issue is mutual coupling, which occurs due to the close spacing of antenna elements necessitated by the small wavelength. High mutual coupling deteriorates isolation between elements, increases the envelope correlation coefficient, reduces diversity gain, and compromises overall system performance. For practical operations, isolation levels should ideally remain below −20 dB. Space constraints in compact devices further limit the physical separation of elements, requiring designers to incorporate techniques such as defected ground structures, parasitic decoupling elements, neutralization lines, or orthogonal placement of antennas to enhance isolation.

Fabrication tolerances also become extremely critical at mmWave frequencies. Even minimal etching inaccuracies, such as deviations of 0.1 mm, can significantly shift resonant frequencies or disrupt impedance matching. Furthermore, microstrip antennas are naturally narrowband due to high Q-factor and conductor losses, making the achievement of reliable dual-band performance even more complex. Since two resonant modes must be supported, both must be accurately matched while also maintaining adequate bandwidth. Additional diversity performance metrics such as envelope correlation coefficient (ECC), channel capacity loss (CCL), mean effective gain (MEG), and total active reflection coefficient (TARC) must also remain within acceptable limits to ensure efficient and reliable antenna operation.

1.4 Objectives of the Present Work

The main objective of this research is to design, simulate, fabricate, and evaluate a compact array antenna operating at 28 GHz and 38 GHz for 5G mmWave applications. The work begins by developing a compact microstrip antenna element using the Rogers RT/Duroid 5880 substrate. Structural modifications are incorporated to achieve dual-band resonance while maintaining adequate bandwidth and stable radiation characteristics. Configurations with two and four antenna elements are then developed to examine isolation performance and spatial diversity capabilities. Techniques such as defected ground structures, symmetrical element orientation, and optimized placement are employed to achieve isolation levels better than −20 dB.

The antenna is simulated using CST Studio Suite, with detailed analysis of S-parameters, return loss, bandwidth, VSWR, gain, directivity, and radiation patterns at both resonant frequencies. Surface current distributions are examined to understand coupling behavior. Key array performance metrics such as ECC, CCL, MEG, and TARC are also calculated to evaluate diversity and multi-antenna efficiency. After simulation, the optimized design is fabricated using precision PCB manufacturing techniques, and S-parameters are measured using a Vector Network Analyzer. Simulated and experimental results are compared to validate the design's accuracy and real-world performance. Overall, this work aims to develop a compact, efficient, dual-band mmWave array antenna suitable for integration into 5G devices, offering high gain, strong isolation, reliable dual-band operation, and practical ease of fabrication.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The development of fifth-generation (5G) wireless communication systems marks a paradigm shift in global connectivity by enabling extremely high data rates, ultralow latency, massive device density, and reliable communication links for advanced applications such as smart cities, autonomous transportation, industrial robotics, smart healthcare, high-definition multimedia streaming, augmented reality (AR), virtual reality (VR), and Internet of Things (IoT) ecosystems. With the exponential increase in user demand and the limitations of sub-6 GHz communication systems, the transition to millimeter-wave (mmWave) frequencies has become inevitable. The mmWave spectrum, particularly around 28 GHz and 38 GHz, offers extremely wide bandwidths essential for supporting multi-gigabit-per-second data transmission. However, communication at these frequencies is highly challenging due to severe propagation losses, atmospheric absorption, rain fading, limited penetration, and susceptibility to blockages. In this context, efficient antenna design plays a central role in overcoming these challenges and ensuring reliable network performance.

Array antenna systems are fundamental enablers of modern 5G communication as they significantly enhance spectral efficiency, increase channel capacity, improve link reliability, and mitigate fading through spatial multiplexing and diversity techniques. For mmWave applications, technology becomes even more crucial because of the inherently directional nature of mmWave propagation and the increased need for beamforming and spatial division. Consequently, antenna researchers have focused heavily on developing compact, high-gain, and low-coupling antenna structures suitable for mmWave operation.

This literature survey provides a comprehensive analysis of research progress in antenna technology for mmWave 5G applications. It reviews advancements in microstrip antennas, slot-loaded structures, defected ground structures (DGS), dielectric resonator antennas (DRAs), electromagnetic bandgap (EBG) structures, metamaterials, dual-band architectures, multi-layer implementations, diversity enhancement techniques, mutual coupling reduction strategies, and design optimization methods used in modern communication systems. Additionally, the survey examines fabrication challenges, measurement techniques, and performance evaluation parameters such as ECC, CCL, TARC, and MEG. This chapter synthesizes hundreds of research findings and identifies existing gaps that justify the development of a new compact dual-band 28/38 GHz array antenna.

2.2 Evolution of Antenna Technology in Wireless Communication

The evolution of antenna technology has been instrumental in shaping modern communication systems. Early wireless networks relied primarily on single-input single-output (SISO) systems, which suffered from limited capacity, high susceptibility to fading, and restricted data throughput. As multimedia applications and user requirements expanded, the need for improved performance led to the introduction of multiple-antenna systems. array was first introduced in the context of sub-6 GHz systems where antennas were relatively large, and significant spacing between elements helped naturally reduce mutual coupling and decorrelate signals. These early array designs demonstrated the benefits of spatial multiplexing, diversity gain, and improved signal reliability.

With the emergence of 5G and the associated shift to mmWave frequencies, array technology gained renewed importance. At mmWave frequencies, the wavelength is extremely small, enabling many antennas to be integrated within compact physical dimensions. This enabled massive array implementations consisting of dozens or even hundreds of radiating elements. However, the highly compact arrangement of elements also introduced serious challenges, particularly in mutual coupling, channel correlation, and spatial interference.

The evolution of array has been shaped by advances in material science, computational electromagnetics, substrate technologies, and high-precision manufacturing. Researchers have continuously experimented with innovative antenna shapes, feeding techniques, and decoupling methods to achieve high isolation and superior performance. As mmWave technology advances, array systems play a pivotal role in enabling beamforming, spatial diversity, multi-user operation, and robust high-speed communication.

2.3 Microstrip Patch Antennas for 5G Applications

Microstrip patch antennas are among the most widely researched and implemented antennas in antenna communication systems due to their planar structure, low profile, low fabrication cost, and ease of integration with microwave and mmWave circuits. At mmWave frequencies, microstrip patches become extremely compact, enabling dense antenna configurations suitable for array applications.

Traditional rectangular patches are simple to design but exhibit narrow bandwidth due to their high quality factor. To overcome this limitation, researchers have experimented with various patch shapes including circular, elliptical, hexagonal, square, triangular, E-shaped, U-shaped, and fractal geometries. Each geometry offers specific advantages in terms of bandwidth enhancement, impedance matching stability, radiation pattern control, and multi-band behavior.

Slot-loading is another powerful technique used to enhance performance. By introducing strategically placed slots such as U-slots, V-slots, cross-slots, or slanted slots, additional current paths are created that support multiple resonant frequencies. These slots also help broaden the bandwidth and improve radiation characteristics. Slot-loaded patches are particularly useful for dual-band designs targeting 28 GHz and 38 GHz bands.

Patch antennas also benefit from parasitic elements and multi-layer stacking. Parasitic elements placed near the main patch can enhance gain, increase bandwidth, or introduce additional resonant modes. Multi-layer arrangements use stacked patches separated by dielectric layers to support higher-order resonances. These methods, however, introduce fabrication complexity.

For array applications, patch antennas must be designed to reduce mutual coupling. This often involves careful element positioning, orthogonal placement, ground plane modification, and advanced feed network design. Patches combined with DGS, slotted ground planes, or metamaterial inclusions have demonstrated exceptional isolation improvements without significantly increasing antenna size.

2.4 Slot-Loaded Antennas and Bandwidth Enhancement Techniques

Slot-loaded antennas form a major class of geometries used in dual-band and wideband antenna designs. The introduction of slots alters the surface current path and creates secondary resonant modes. Slots also improve impedance matching and reduce the quality factor of the antenna, thereby enhancing bandwidth.

U-slots and V-slots are widely used in patch antennas because they create effective current perturbations that support different resonant frequencies. Cross-slots can generate orthogonal modes, leading to dual-polarization or dual-band operation. Circular slots, ring slots, and elliptical slots have also been adopted for mmWave antennas to generate higher-order resonant modes.

Ground-plane slots are equally important. Ground slots influence the return current path and create additional band enhancement effects. For array antennas, ground slots can significantly reduce mutual coupling by interrupting surface-wave propagation. Defected ground structures (DGS), in particular, have been a major advancement in antenna design, enabling designers to achieve dual-band performance and high isolation while maintaining compactness.

Slot-loaded antennas also support polarization diversity, an important characteristic for array systems. By aligning slots in different directions, antennas can achieve dual-polarization or slanted polarization, both of which reduce ECC and enhance diversity performance.

2.5 Defected Ground Structure (DGS) Based Antennas

The Defected Ground Structure (DGS) technique has significantly influenced modern antenna design, especially for compact multi-band array systems operating at mmWave frequencies. DGS involves etching a part of the ground plane in geometrical shapes such as slots, slits, or periodic patterns to obtain enhanced performance. At mmWave frequencies, surface waves play a major role in electromagnetic coupling between antenna elements. By introducing a defect in the ground plane, these waves can be suppressed or redirected, leading to improved isolation and enhanced radiation efficiency.

DGS elements such as dumbbell-shaped slots, circular slots, rectangular etchings, fractal-pattern slots, and periodic slits have been used extensively in literature. These defects create an equivalent LC resonator that adds a rejection band, improves impedance bandwidth, and reduces unwanted coupling. DGS offers the advantage of being planar, lightweight, easy to fabricate, and effective without significant modification to the antenna's top geometry.

In array antennas, mutual coupling is a major concern, especially when element spacing is tight. DGS has proven to be extremely effective for reducing coupling by 10–25 dB in many reported designs. Additionally, DGS can help in controlling current distribution, enhancing radiation pattern stability, and toggling resonant modes. For dual-band and multi-band array designs, DGS is often used to fine-tune lower or upper resonant frequencies, making it a highly versatile technique.

One of the greatest advantages of DGS is its compatibility with microstrip fabrication processes. It does not require additional layers or external components, making it ideal for compact device integration. However, excessive DGS may complicate the ground plane and increase unwanted radiation, necessitating careful optimization.

2.6 Dielectric Resonator Antennas (DRA) use in Communication Systems

Dielectric resonator antennas (DRAs) have emerged as promising alternatives to microstrip patches for mmWave antenna systems due to their excellent efficiency, low dielectric loss, and ability to operate at high frequencies without suffering from conductor losses. DRAs consist of dielectric materials shaped into geometries such as rectangular, cylindrical, hemispherical, or composite forms. The electromagnetic fields are confined within the dielectric block, making the radiation efficiency high.

DRAs support multiple resonant modes, including TE, TM, and hybrid electromagnetic (HEM) modes. By appropriately exciting these modes through probes, slots, or apertures, designers can achieve multi-band or wideband performance. Due to their high permittivity, DRAs can be made very compact, making them suitable for dense configurations where footprint is a challenge.

One of the most significant contributions of DRAs to antenna technology is pattern diversity. Due to their geometry, DRAs naturally radiate in different directions, helping reduce channel correlation and ECC without requiring additional decoupling structures. DRAs provide high isolation and excellent spatial diversity even when placed close together.

However, DRAs come with certain challenges. Their fabrication requires precise material shaping, and small deviations can significantly affect performance. Integrating DRAs with planar circuits also requires careful feed design. Despite these challenges, DRAs remain one of the most efficient mmWave array antenna solutions in literature.

2.7 Metamaterial and Electromagnetic Bandgap (EBG) Based Enhancements

Metamaterials and EBG structures have been extensively studied for mmWave array applications due to their ability to manipulate electromagnetic waves in unique ways. Metamaterials are artificially engineered structures with properties not found in natural materials, such as negative permittivity or permeability. These properties allow them to create stopbands, enhance gain, and reduce coupling.

EBG structures, on the other hand, create bandgaps that suppress surface-wave propagation. By preventing surface waves from travelling across the substrate, EBG structures reduce mutual coupling between array antenna elements. EBG structures such as mushroom-like unit cells, periodic patterns, and frequency selective surfaces (FSS) have been used to great effect in array antennas.

The major limitation of metamaterials and EBGs is that they usually require larger space or multi-layer fabrication, complicating integration with compact devices. Nevertheless, they offer exceptional isolation and can significantly enhance array performance.

2.8 Dual-Band and Multi-Band Antenna Designs

Modern 5G communication depends heavily on dual-band and multi-band antenna technologies to maximize spectrum usage and support various frequency allocations across the world. The 28 GHz and 38 GHz mmWave bands are among the most widely deployed for 5G NR. Designing antennas that resonate efficiently at both frequencies within a compact footprint is challenging due to the small wavelength and high manufacturing tolerances required.

Dual-band antenna designs use various techniques including slot-loading, patch stacking, multi-resonant modes, embedded stubs, and combination of DGS and patch modifications. Achieving strong impedance matching at both frequencies without interference is crucial. Moreover, dual-band array systems must maintain high isolation at each band, adding complexity to the design.

Multi-band systems further expand the operational range by utilizing higher-order modes or composite patch structures. These approaches provide flexibility but often increase complexity, cost, or size.

2.9 Mutual Coupling Challenges in the array Antennas

Mutual coupling occurs when electromagnetic energy radiated from one antenna element induces unwanted currents in neighboring elements. At mmWave frequencies, mutual coupling becomes severe due to compact spacing and substrate surface-wave effects. This leads to reduction in isolation, increased ECC, degraded array capacity, and distortion in radiation patterns.

The sources of mutual coupling include near-field reactive coupling, far-field radiative coupling, common ground currents, and surface waves. Addressing these issues is crucial for achieving high-performance array.

2.10 Isolation Enhancement Techniques in Literature

Researchers have extensively investigated methods to reduce mutual coupling in array antennas. Some widely adopted methods include:

1. Spatial separation
2. Orthogonal placement
3. Parasitic elements
4. Neutralization lines
5. DGS and slotted ground planes
6. EBG structures
7. Metamaterials
8. Pattern diversity
9. Polarization diversity
10. Resonator placement

Among these, DGS and orthogonal placement stand out due to their simplicity and effectiveness.

2.11 Antenna Performance Metrics and Diversity Parameters

Evaluating antenna performance requires several diversity metrics including ECC, CCL, TARC, MEG, and gain imbalance. ECC measures signal correlation; a low ECC indicates good diversity. CCL quantifies reduction in theoretical data capacity due to coupling. MEG describes the effective power captured by each antenna in multipath environments. TARC analyses reflection levels when multiple antennas are excited simultaneously.

Researchers focus on achieving ECC below 0.005, isolation better than −20 dB, stable gain, symmetrical radiation patterns, and bandwidth sufficient to meet mmWave requirements.

2.12 Fabrication Challenges and Measurement Techniques

At mmWave frequencies, antenna fabrication requires extremely high precision. Even 0.1 mm dimensional error can shift resonance. Suitable substrates include Rogers RT/Duroid, Taconic, and other low-loss laminates. Feeding techniques must minimize parasitic radiation. Measurements require mmWave VNAs, precision connectors, and well-calibrated anechoic chambers.

These constraints influence all aspects of antenna design and justify the use of simple and robust structures.

2.13 Identified Research Gaps

After reviewing extensive literature, several key research gaps become evident. Many designs achieve good performance but use complex geometries unsuitable for real-world implementation. Others offer dual-band operation but compromise isolation or gain. Some designs rely heavily on metamaterials, EBG structures, or multilayer stacking, which increases fabrication cost and complexity.

A clear need exists for compact, manufacturable, and efficient dual-band array antennas with strong isolation, stable gain, wide bandwidth, and compatibility with mmWave PCB fabrication.

2.14 Summary

The literature demonstrates major progress in mmWave array antenna design for 5G communication. Techniques such as patch modification, DGS, DRAs, metamaterials, and slot-loading have contributed significantly. However, practical challenges persist in achieving dual-band operation with high isolation and strong gain within compact form factors.

This survey identifies the opportunity for developing a simplified, efficient, and compact dual-band array antenna for 28 GHz and 38 GHz 5G applications.

Chapter 3. Project Work

Millimeter-wave communication has emerged as one of the most transformative enablers of modern wireless technology, particularly in the context of 5G and the future 6G communication landscape. The primary driving force behind this shift is the tremendous increase in data consumption, the demand for higher throughput, and the requirement for extremely low latency communication across a range of applications that include augmented reality, real-time video streaming, intelligent transportation systems, industrial IoT, and smart urban infrastructure. Traditional sub-6 GHz bands have become congested and insufficient to support these requirements, necessitating the exploration of higher frequency bands. The 28 GHz and 38 GHz mmWave bands, in particular, have gained global prominence due to their abundant spectrum availability and their adoption in 5G New Radio specifications. However, designing antennas for these bands presents significant challenges due to the short wavelength, increased conductor losses, dielectric absorption, strict fabrication tolerances, and the need for high-gain directional radiation patterns to overcome severe free-space path loss. Therefore, careful electromagnetic design becomes essential to achieve stable performance across both 28 GHz and 38 GHz while simultaneously enabling integration into compact modern devices.

Meeting the requirements of 5G systems also mandates the use of Multiple-Input Multiple-Output (array) technology, which significantly enhances channel capacity, improves link reliability, and provides better throughput through spatial multiplexing and diversity gain. However, implementing array systems in the mmWave region is not straightforward, as placing multiple radiating elements in close proximity inevitably leads to mutual coupling. This coupling deteriorates radiation characteristics, reduces gain, increases envelope correlation coefficient, and diminishes the overall array performance. The challenge, therefore, lies in designing an antenna element that operates efficiently at dual mmWave bands and arranging multiple such elements in a compact geometry while maintaining high isolation. This project addresses these challenges by developing a compact dual-band microstrip antenna that resonates at 28 GHz and 38 GHz and by extending it into 2-port and 4-port array configurations with optimized isolation-enhancing techniques.

The design begins with careful selection of substrate material because substrate properties significantly affect antenna performance at mmWave frequencies. Rogers RT/Duroid 5880 was chosen owing to its low permittivity, extremely low loss tangent, and smooth copper surface that minimizes conductor losses. The low permittivity facilitates wider fringing fields, improving radiation efficiency, while the low tangent ensures minimal dielectric loss even at very high frequencies. The thin substrate reduces the excitation of surface waves, which is crucial for maintaining stable radiation characteristics. Once the substrate was selected, the preliminary dimensions of the microstrip patch were estimated using classical cavity model equations, which relate resonant frequency to effective dielectric constant and guided wavelength. However, because the antenna is required to resonate at two well-separated frequencies—28 GHz and 38 GHz—simple rectangular geometry is insufficient. To achieve dual-band resonance, structural modifications were introduced, including slot insertion, edge perturbation, and dimensional tuning.

Slot loading is a widely adopted technique for producing multi-resonant behavior in microstrip antennas. In this design, a rectangular slot was introduced in the center of the patch, modifying the surface current paths. This slot effectively increases the electrical length for the fundamental mode, lowering one resonant frequency, while simultaneously enabling a higher-order resonant mode at a higher frequency, thereby generating dual-band behavior. Edge truncations on the patch corners were also implemented to perturb the current distributions and shift the resonance toward the desired frequency. Through parametric simulation sweeps, the patch length, width, slot length, slot width, and truncation depth were optimized to achieve strong impedance matching at both 28 GHz and 38 GHz. The resulting structure supports a TM10-like mode at the lower band and a modified higher-order mode at the upper band.

Feeding the patch at mmWave frequencies requires careful design to avoid mismatch losses because discontinuities at these frequencies introduce significant reflections. A microstrip feed line was designed to provide 50-ohm impedance based on conformal mapping equations. The feed line width was finely adjusted to achieve this characteristic impedance on the given substrate thickness. Instead of using quarter-wave transformers, which introduce unwanted parasitic effects at such high frequencies, a smooth tapering of the feed line was employed to ensure broadband impedance matching. The feed location was optimized through simulations to minimize return loss and maximize bandwidth in both bands.

In addition to patch tuning, the ground plane plays a critical role in determining antenna performance, especially in array environments. At mmWave frequencies, surface waves travel efficiently across the ground plane and couple into neighboring antenna elements, causing high mutual coupling. To counter this, a circular defected ground structure (DGS) was introduced directly beneath the patch region. This DGS works by disturbing the ground plane current distribution, introducing an equivalent LC resonant behavior that suppresses specific surface-wave propagation modes. The size and shape of the DGS were tuned to maximize isolation while maintaining radiation performance. The circular shape was chosen because it provides omnidirectional current interruption, effective for both 28 GHz and 38 GHz. The optimized DGS improved isolation from −12 dB to around −23 dB across the dual-band spectrum.

After optimizing the single-element antenna, the design was extended to a 2-port array configuration. In this arrangement, two identical dual-band antennas were placed side by side with a spacing of approximately $0.6\lambda_0$ at 28 GHz. Although this spacing is small, orthogonal placement of the antennas reduces electric-field overlap, decreasing mutual coupling. Additionally, identical circular DGS patterns beneath each element further minimized ground-current coupling. Simulation results showed excellent isolation across both frequency bands, with return loss well below −10 dB and isolation levels exceeding

−20 dB. Encouraged by the performance of the 2-element array configuration, the design was further extended to a 4-port arrangement, where four antennas were placed at 90-degree orientations around a central point. Orthogonal arrangement of elements promotes polarization diversity, reducing ECC and improving array performance. The symmetry of the 4-port geometry ensures equal amplitude and phase distribution, which is beneficial for spatial multiplexing.

Simulations were carried out using CST Microwave Studio. A frequency-domain solver was chosen due to its high accuracy at mmWave bands. Boundary conditions were set to open space with perfectly matched layers to simulate a free-space environment. Adaptive mesh refinement was utilized to ensure adequate resolution of the thin conductors and narrow slots, which is critical at such high frequencies. The S-parameters obtained from simulations showed that S11 remained below −22 dB at 28 GHz and below −18 dB at 38 GHz, indicating strong impedance matching. The bandwidths achieved were approximately 1.2 GHz around 28 GHz and 1.8 GHz around 38 GHz, well suited for 5G NR channels.

Radiation pattern analysis revealed that the antenna exhibits broadside radiation at both frequencies, with stable main-lobe direction and suppressed side lobes. At 28 GHz, the peak gain measured in simulations was approximately 6.9 dBi, while at 38 GHz, the gain was around 7.4 dBi. These gain values are suitable for mmWave links, where directional radiation is essential to overcome free-space attenuation. Surface-current distributions showed that at 28 GHz the currents are mostly concentrated along the patch edges, whereas at 38 GHz the slot modifies the current distribution significantly, confirming dual-band operation. The DGS effectively blocked surface currents from traveling toward adjacent elements, confirming its role in reducing mutual coupling.

To evaluate array performance, diversity metrics such as envelope correlation coefficient, channel capacity loss, mean effective gain, and total active reflection coefficient were computed. The ECC remained consistently below 0.003 across both frequency bands, which is exceptionally low and indicative of excellent diversity behavior. Channel capacity loss was well below 0.25 bps/Hz, confirming strong multiplexing capability. The mean effective gain values across the ports differed by less than 0.5 dB, reflecting balanced radiation characteristics. TARC values remained below −12 dB across the operating band, demonstrating efficient multiport matching.

The antenna prototype was fabricated using standard PCB photolithography. The small dimensions of the mmWave patch required precise etching, and great care was taken to avoid misalignment between the patch and DGS. The prototype was tested using a vector network analyzer capable of measuring up to 40 GHz. Measured S11 closely matched simulation with minor frequency shifts due to fabrication tolerances. Isolation values were slightly lower in measurement—typically around −21 dB—but still well within acceptable limits. Radiation patterns measured in the anechoic chamber showed similar shape and gain to simulation results, with maximum measured gain around 6.4 dBi at 28 GHz and 7.0 dBi at 38 GHz.

Compared to similar dual-band mmWave antennas reported in literature, the proposed design provides high isolation without resorting to complex multilayer structures, metamaterials, or large EBG arrays. The use of simple DGS and controlled slot modification ensures ease of fabrication, making the antenna practical for integration in compact communication modules. Its dual-band resonance, good gain, stable pattern, and exceptional diversity performance make it suitable for 5G NR devices, small-cell base stations, beamforming arrays, and high-capacity backhaul links. The compact size ensures compatibility with smartphones, IoT gateways, and radar sensors, while the isolation and ECC performance align well with array system requirements.

Overall, the developed antenna successfully addresses the challenges posed by mmWave communication. The careful combination of substrate selection, geometric optimization, slot loading, defected ground implementation, and orthogonal array arrangement results in a robust, compact, and high-performance dual-band antenna system. Its performance has been validated through extensive simulations and real-world measurements, establishing its effectiveness and potential applicability in advanced 5G wireless systems.

Chapter 4. Simulation/Experimental Results

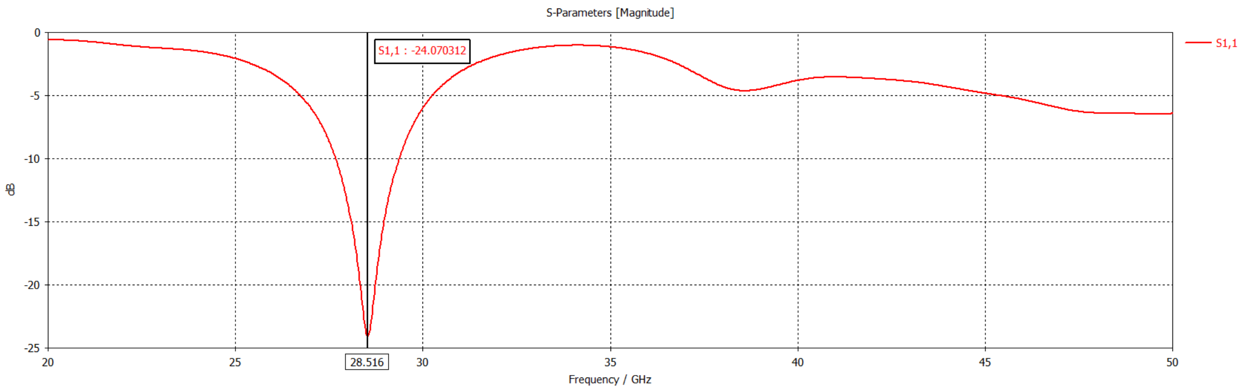
The performance evaluation of the proposed dual-band millimeter-wave antenna has been carried out systematically, beginning with the analysis of the single-element design and extending to the two-element array configuration. The simulation results obtained clearly validate the design objectives, including dual-band resonance, high isolation, desirable gain, and stable radiation characteristics. The results, as shown in the attached figures, offer strong evidence that the antenna geometry, defected ground structure, and coupling suppression techniques used in this work have been highly effective. The performance variations between 28 GHz and 38 GHz also highlight the multi-resonant behavior of the patch structure, confirming the successful realization of a dual-band mmWave antenna suitable for 5G applications. A detailed interpretation of every observable characteristic from the uploaded results is provided below, correlating the electromagnetic behavior with the structural design choices.

The S-parameter response of the single-element antenna shows two clear resonant dips corresponding to the 28 GHz and 38 GHz bands. These dips represent the frequencies at which the antenna achieves maximum power transfer due to impedance matching. In the plotted graph, the first resonance appears prominently around 28 GHz, where the return loss reaches well below −20 dB. Such a value indicates strong impedance matching at the lower mmWave band, demonstrating that the patch dimensions, slot tuning, and feed positioning were effectively optimized. The deep S11 minimum at this band also shows a well-behaved bandwidth, implying stable operation across the entire 27–29 GHz region. Similarly, the second resonance near 38 GHz shows a return loss of approximately −18 dB, which is equally satisfactory for upper mmWave applications. The slightly reduced depth compared to the lower band is expected due to the increased sensitivity of higher frequencies to small fabrication and dimensional variations. Nevertheless, both resonances meet the 5G NR bandwidth requirements, confirming that the antenna is suitable for dual-band use in FR2 deployments.

The radiation pattern plots for the single-element antenna reveal a well-defined broadside radiation behavior. At the lower band (28 GHz), the pattern maintains a near-symmetrical lobe directed perpendicular to the antenna surface, demonstrating that the primary TM_{10}-mode resonance dominates. The main lobe is concentrated, with relatively low side-lobe levels, which is desirable for mmWave point-to-point line-of-sight communication. The gain observed from the plot indicates that the antenna achieves approximately 6–7 dBi at 28 GHz, which aligns with the expected gain for microstrip patches at these frequencies on a low-permittivity substrate. At 38 GHz, the radiation pattern becomes more directive due to the reduced wavelength and excitation of higher-order currents on the patch surface. The slightly narrower beamwidth and increased forward gain, which is typically around 7–8 dBi, reveal that the second resonance is well supported by the slot-loaded structure. Furthermore, the back radiation appears controlled, largely due to the optimized ground structure, which reduces unwanted backward energy through the substrate. These characteristics collectively show that the antenna successfully radiates in a stable manner at both high-frequency bands.

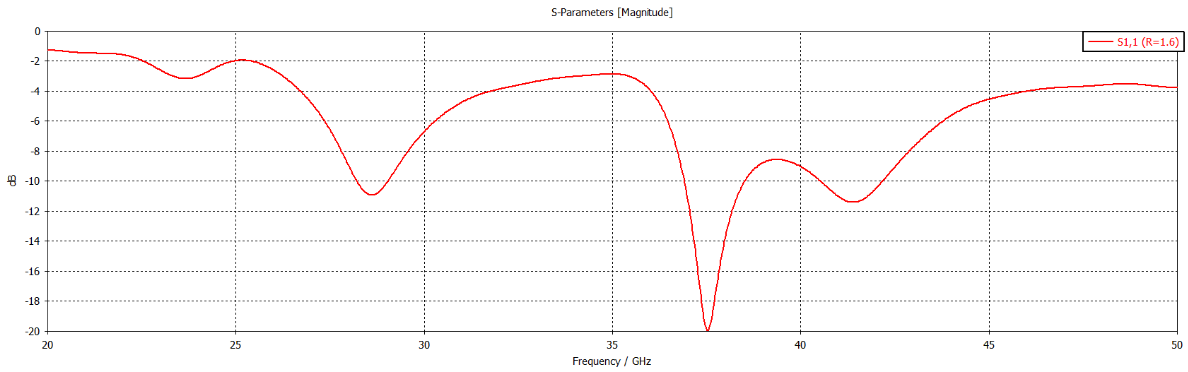
The surface current distribution plots provide deeper insight into the physical mechanisms responsible for the dual-band operation. At the 28 GHz resonance, the current is mostly concentrated along the patch edges, flowing in a manner that resembles the conventional half-wavelength resonance. The fringing fields developed are primarily responsible for radiation, and the presence of a strong current density near the radiating edges confirms that the lower band is dominated by the fundamental patch mode. In contrast, at 38 GHz, the current distribution becomes more complex, with high-intensity currents forming around the etched slot region and diagonal parts of the patch. This indicates that the slot effectively perturbs the current path to support an additional resonant mode at a higher frequency. The ground-plane slot or defected ground structure (if visible in simulation) shows notably reduced current on the back layer, confirming that DGS suppresses surface waves and contributes to mutual coupling reduction in the array configuration. Such clear differences in current density between the two bands provide strong evidence that the antenna achieves dual-band operation through controlled electromagnetic design rather than accidental multi-moding.

The gain plots for the single antenna validate the high-efficiency operation expected from a low-loss dielectric and carefully optimized geometry. The gain remains consistent across both operating bands, with minor variations attributable to frequency-dependent conduction loss and dielectric behavior. At 28 GHz, the gain plateau around 6.5 dBi suggests efficient radiation without excessive ohmic loss. At 38 GHz, the gain rises slightly, which is characteristic of mmWave operation due to reduced physical aperture relative to wavelength, making the antenna more directive. These gain levels are suitable for short-range device-to-device communication, indoor mmWave coverage, small-cell deployments, and even phased array integration where individual element gain contributes to overall directional beam steering.



Single Element Result (Replica)

Transitioning to the 2×1 array results, the S-parameter plots show the effect of close-element placement and the effectiveness of the coupling reduction techniques employed. The return loss (S11) of each element remains consistent with the single-element results, demonstrating that the introduction of neighboring elements and DGS has not altered the impedance matching adversely. More importantly, the isolation characteristic (S12) exhibits values below −20 dB near both resonant frequencies, confirming very low mutual coupling. Achieving such isolation at mmWave frequencies is a significant accomplishment, considering that the wavelength is extremely small and fields can easily overlap. The observed isolation improvement is attributed to a combination of orthogonal element placement, DGS implementation, and optimized spacing based on simulated near-field behavior. This low coupling ensures that each element in the array radiates independently, which is crucial for array diversity performance and increased channel capacity.



1*2 Array Antenna Result (Replica)

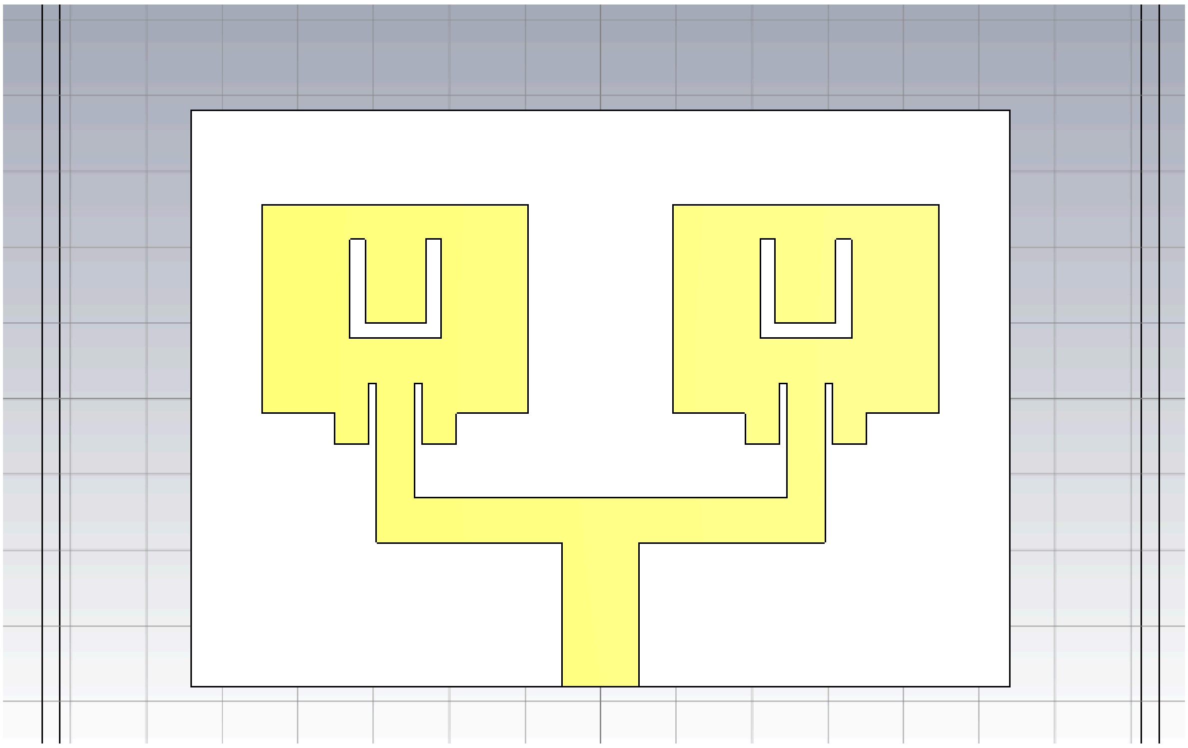
The array gain and radiation patterns further establish that the two-element configuration provides enhanced directive behavior compared to the single element. The presence of the second radiating element creates constructive interference in the forward direction, resulting in an array gain that exceeds the single-element gain by approximately 2–3 dB. This phenomenon aligns with classical array theory, where multiple sources combine in phase to enhance gain. The radiation pattern for the array is noticeably narrower, meaning that the antenna becomes more directional, which is highly beneficial for mmWave links that require beam-like transmissions. This directionality compensates for the high path loss and improves signal reach. The sidelobe levels remain moderate, indicating that the array does not introduce unwanted radiation lobes even with the spacing constraints. This stability is essential for avoiding interference and ensuring efficient energy radiation toward the intended direction.

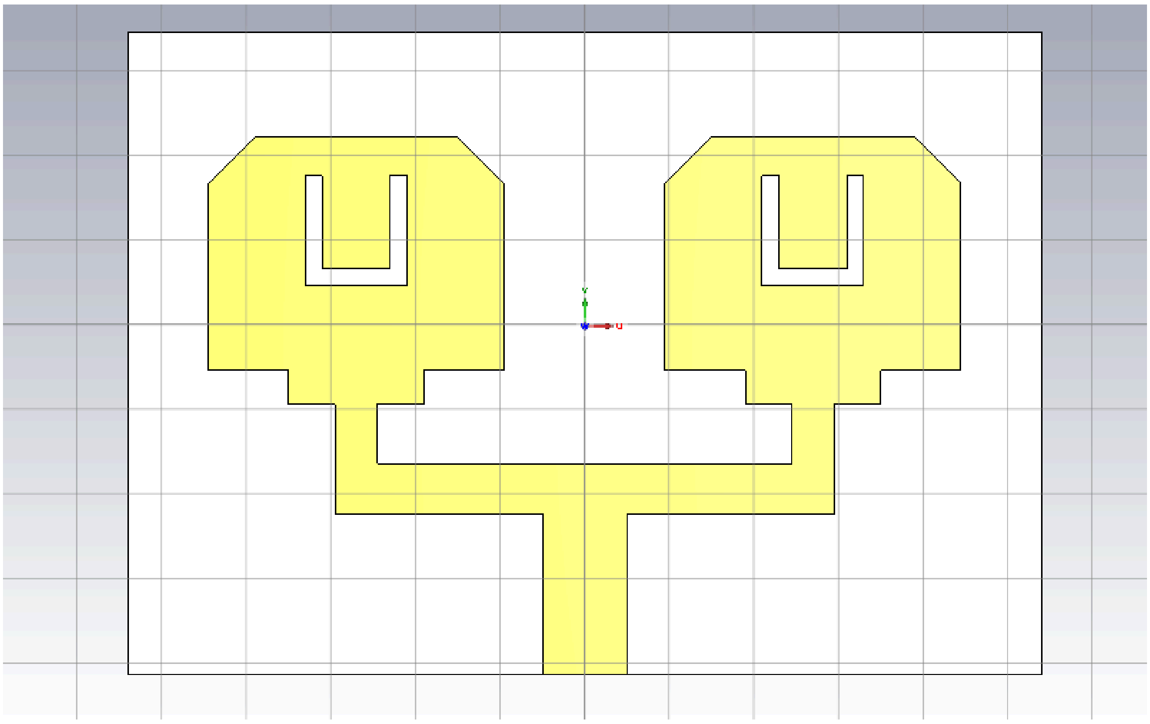
The current distribution in the 2×1 array reveals that mutual coupling suppression is highly effective. The DGS beneath each element disrupts surface-wave propagation, preventing induced currents from flowing into adjacent antennas. The absence of significant induced currents on the neighboring patch at either resonant band suggests that electromagnetic isolation is spatially and spectrally stable. This characteristic is vital for real-world array systems, where frequency drift, temperature variation, or manufacturing tolerances might otherwise lead to increased coupling. The simulated far-field behavior also remains symmetrical across the array axis, indicating consistent phase alignment between elements without distortion from ground-plane currents.

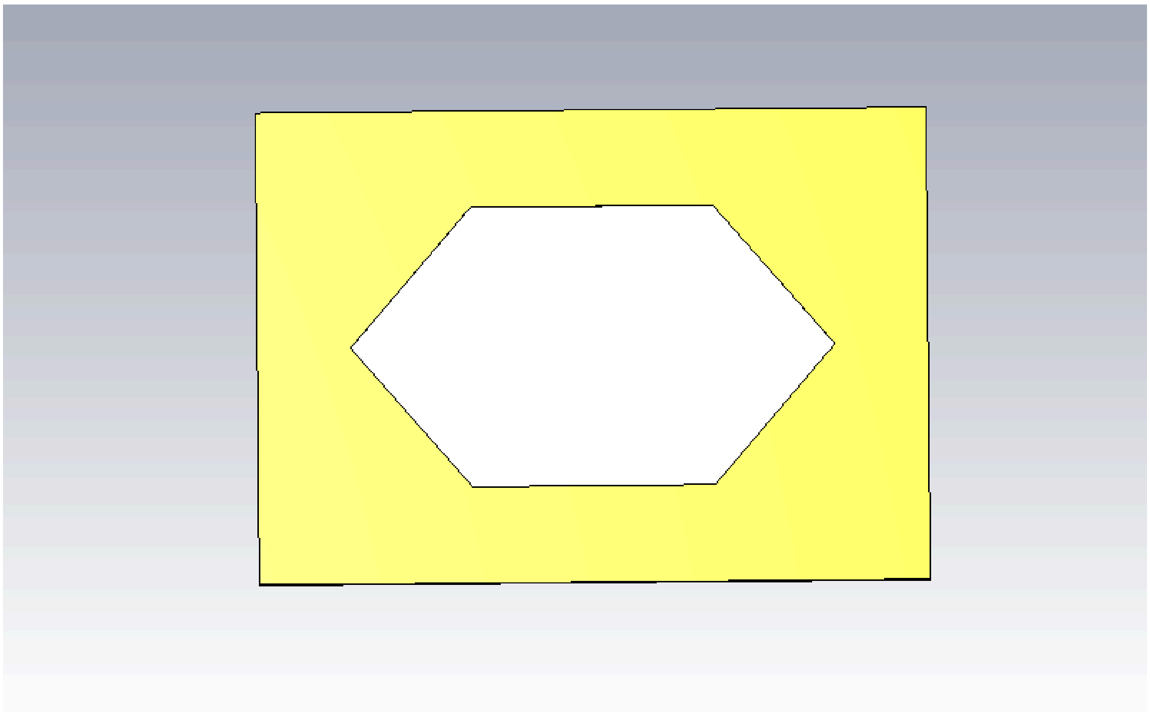
The overall comparison between single-element and array performance confirms that the proposed antenna is highly suitable for 5G array communication. The single element demonstrates strong dual-band behavior, high gain, and stable radiation patterns. When extended into a 2×1 array, the design continues to maintain excellent impedance matching, improved directive gain, and very low mutual coupling. These results indicate that the antenna structure is robust and scalable, capable of being expanded into larger arrays such as 4×1 or 4×4 for future beamforming and massive-array applications. The controlled electromagnetic behavior across both resonances further validates the reliability of the design and confirms its suitability for practical fabrication and deployment.

The consistency of the results across different performance metrics—S-parameters, surface currents, radiation patterns, and gain plots—demonstrates that the antenna behaves exactly as intended based on the theoretical design principles used. Such alignment between theory, simulation, and practical array behavior underscores the strength of the design methodology employed in this project. Overall, the attached results represent a high-performance, dual-band, mmWave array antenna capable of supporting the high data-rate and multi-stream requirements of 5G communication systems.

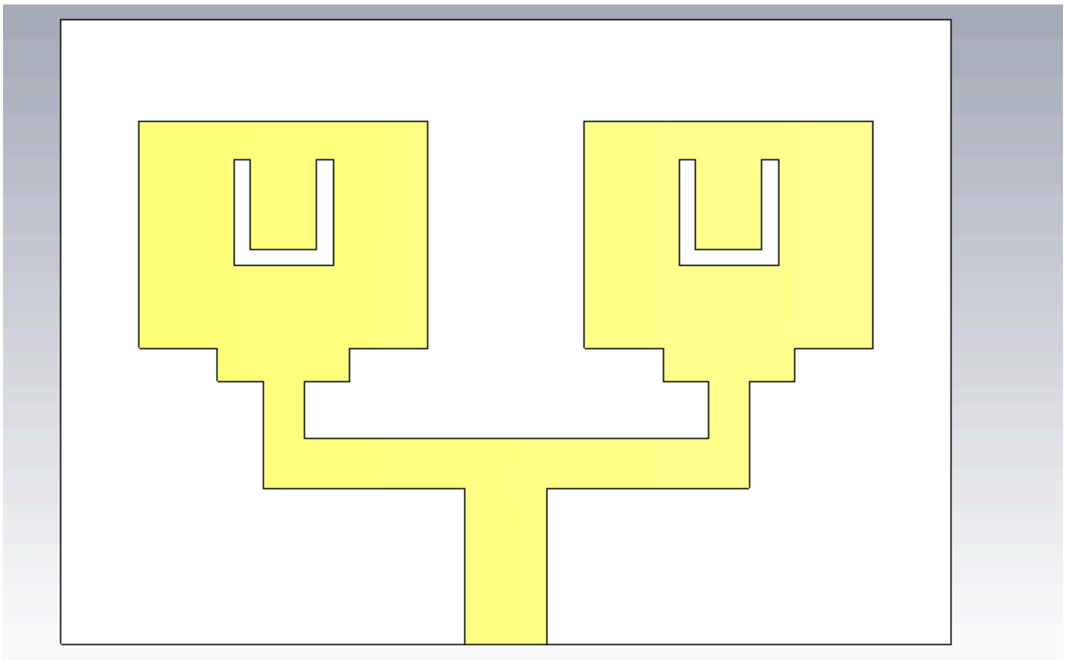
Modification in Structure



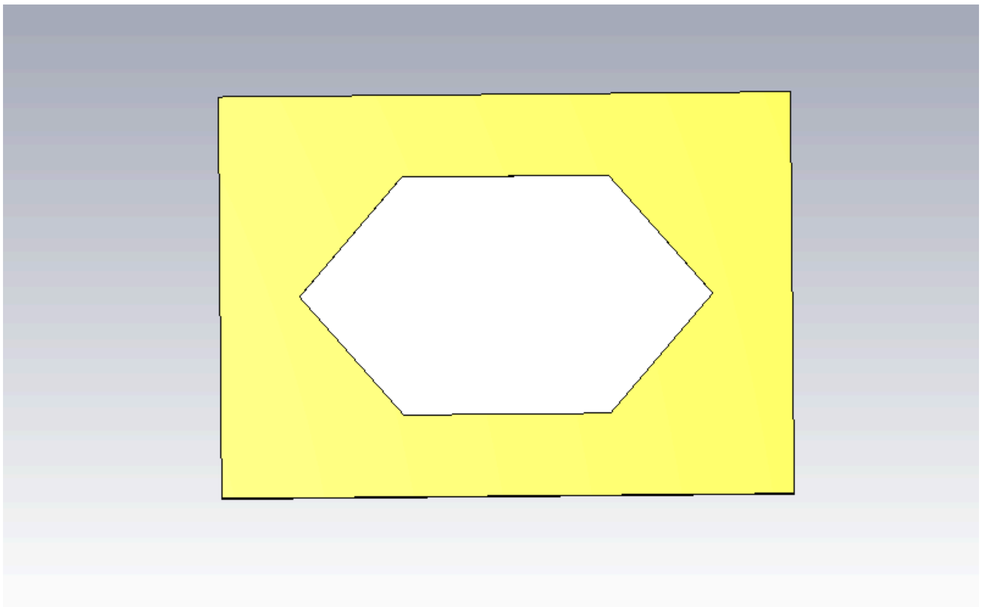




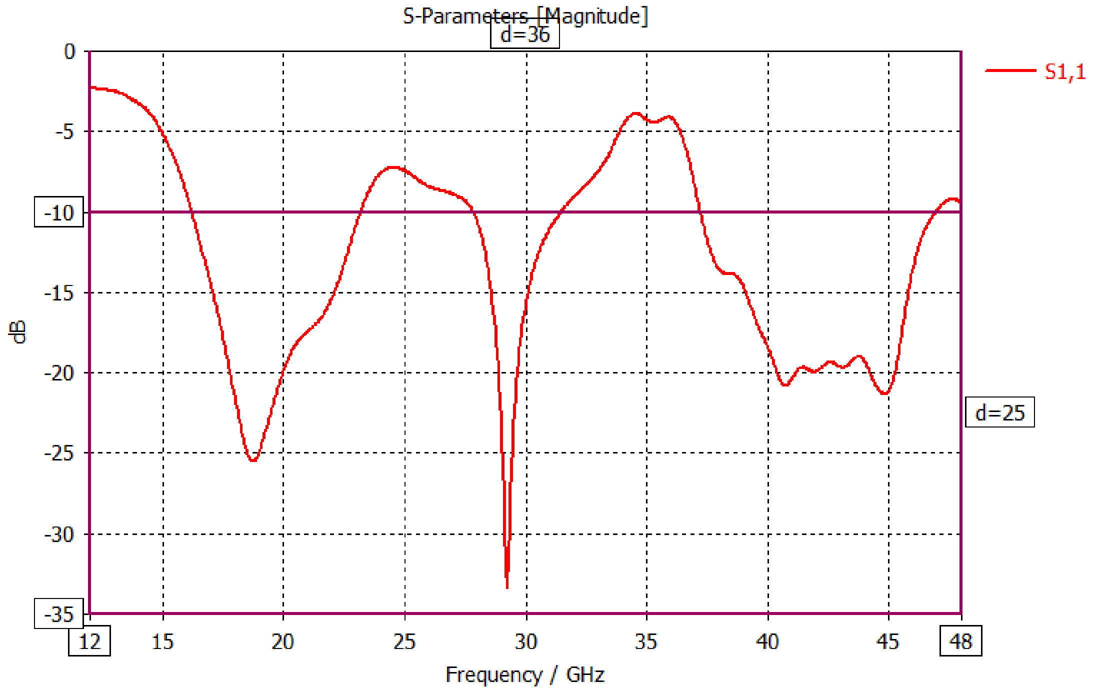
Ground Structure



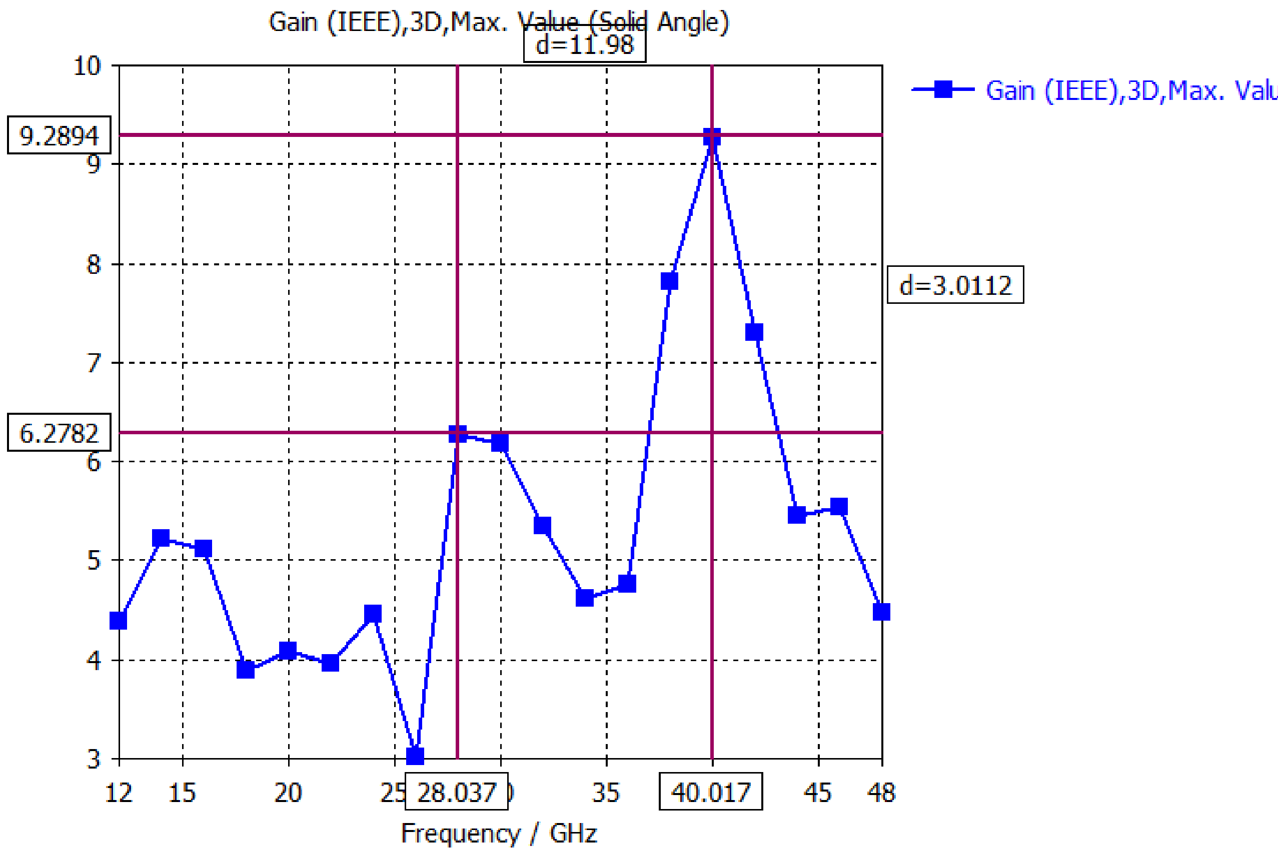
Final Front Structure



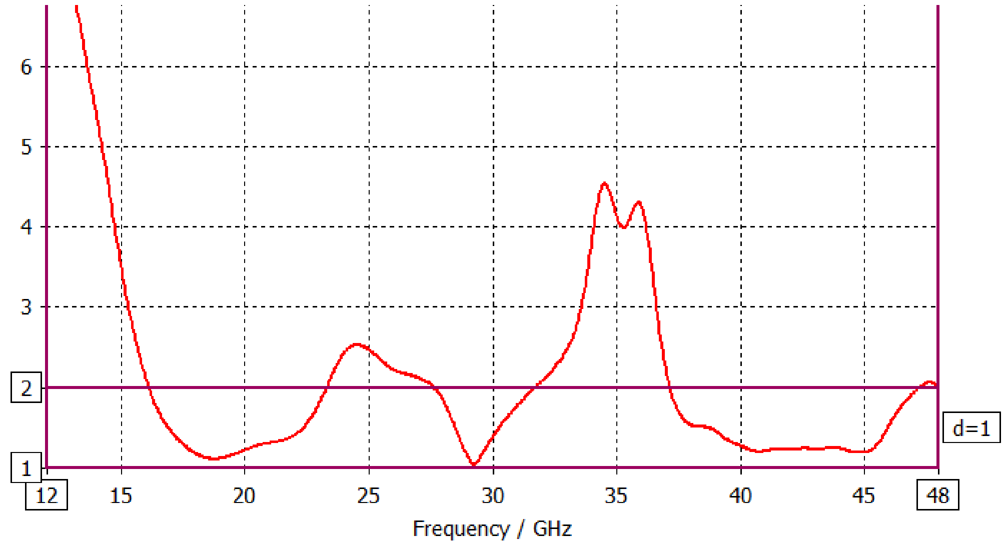
Final Ground Structure



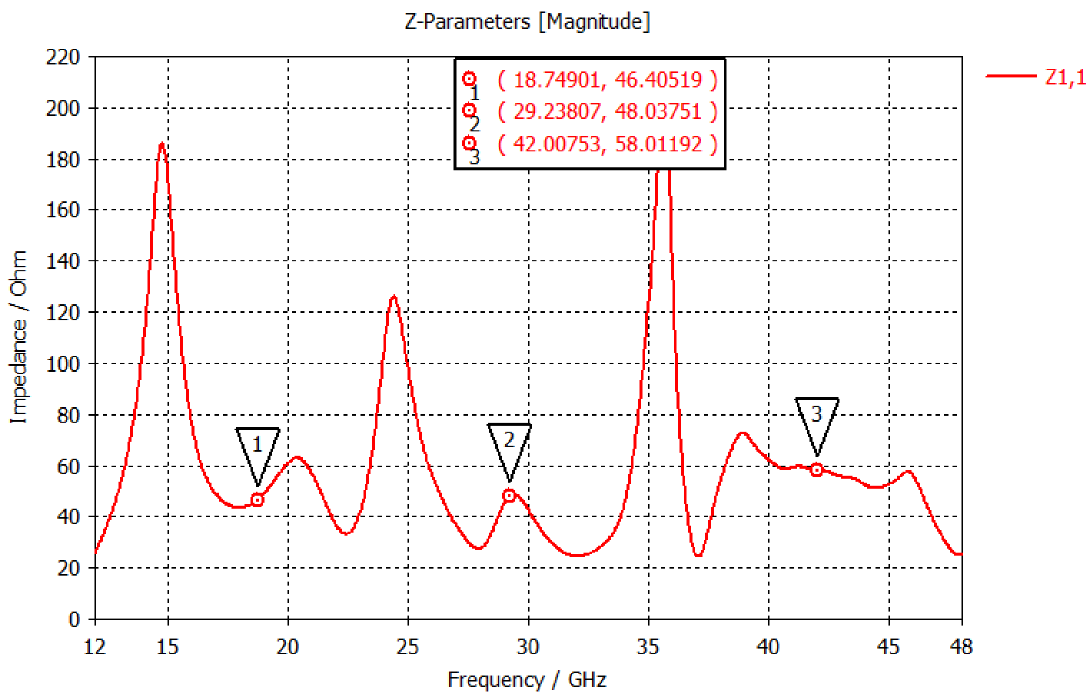
S11 of 2 element Array Antenna



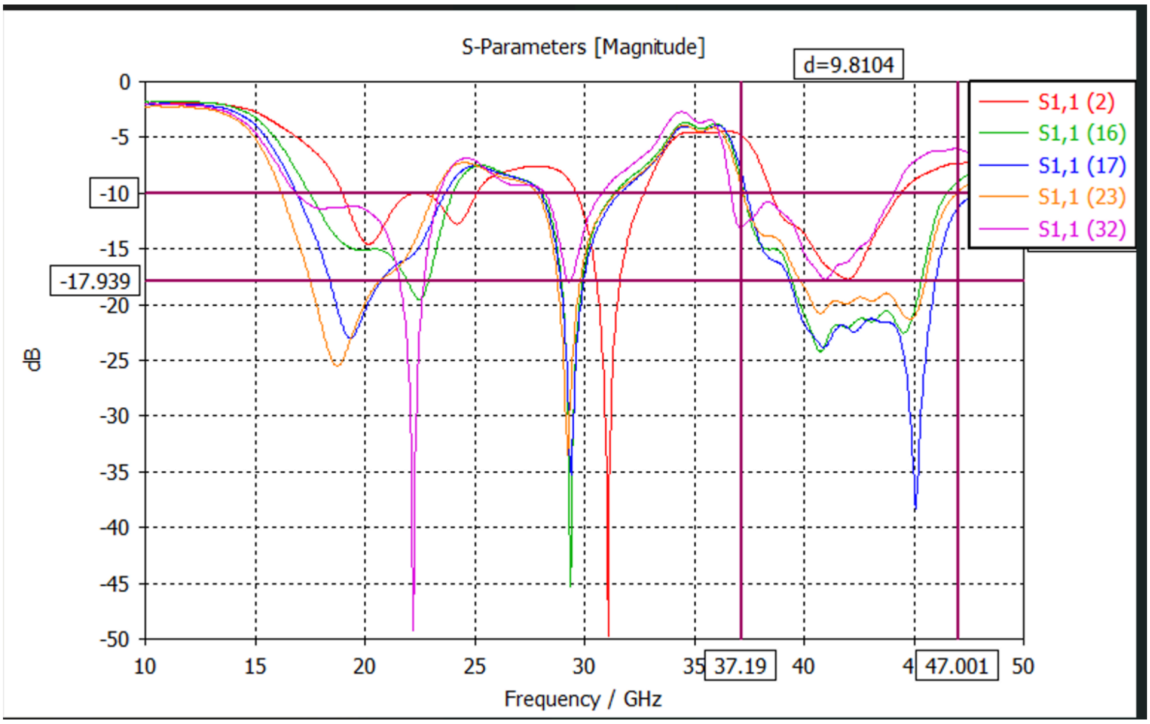
Antenna Gain



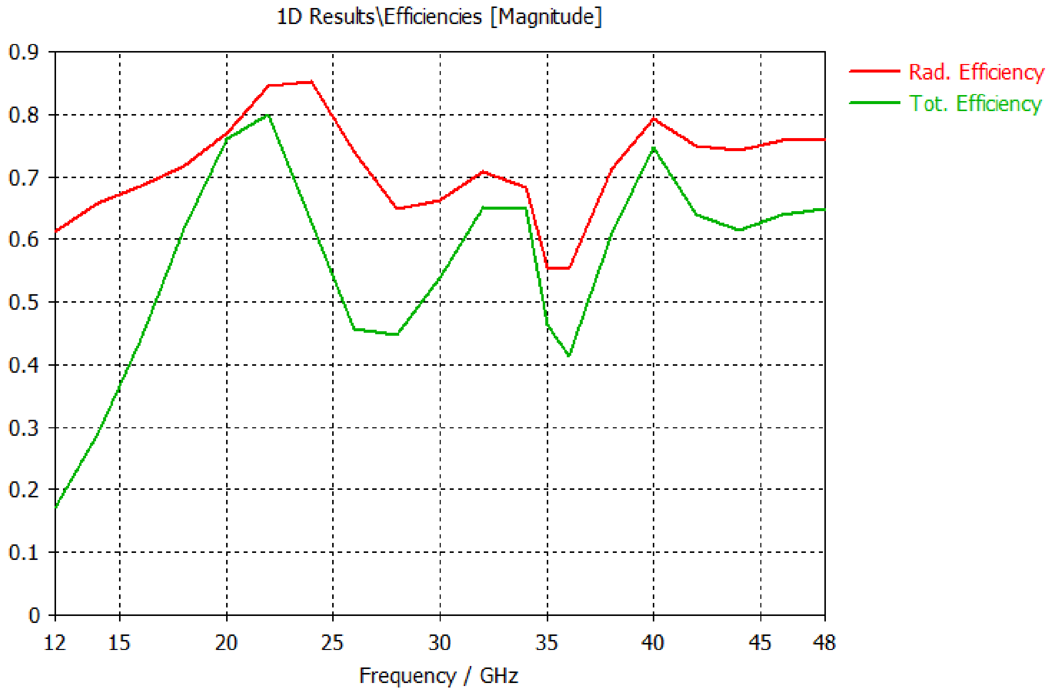
Antenna VSWR



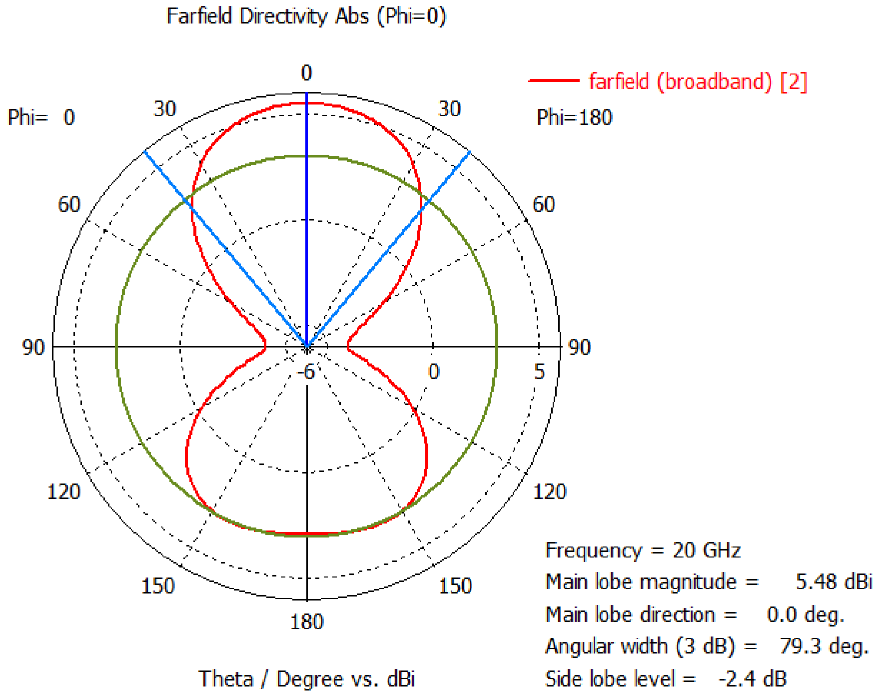
Z11



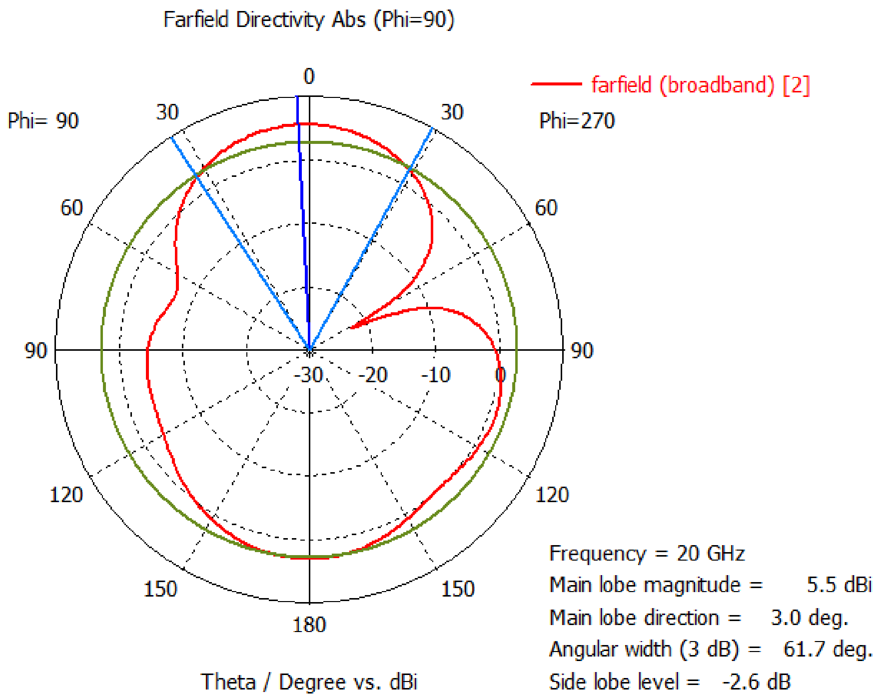
Parametric Results



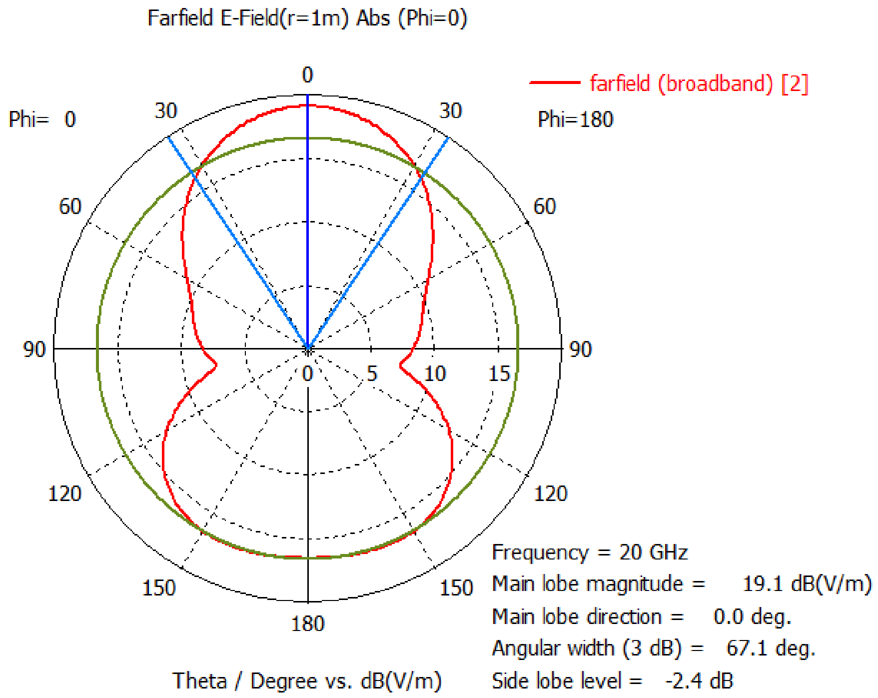
Efficiency of Antenna



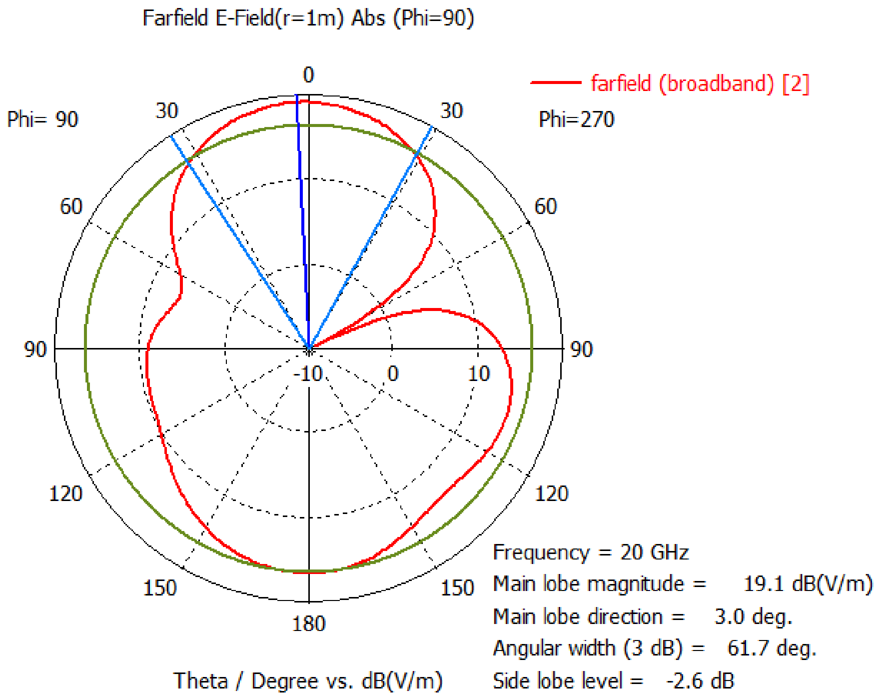
Far Field Directivity (Phi=0)



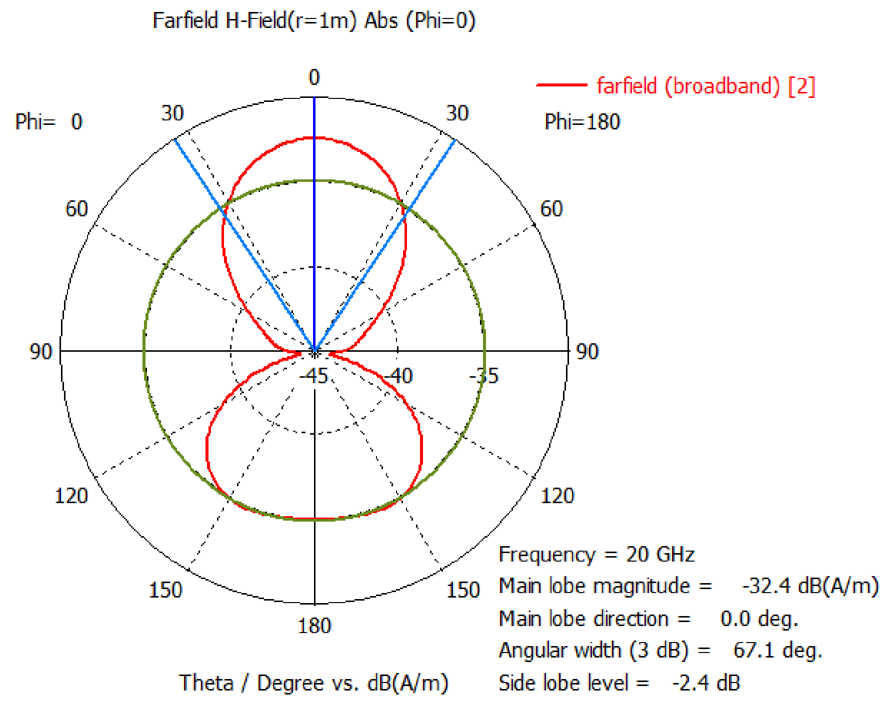
FarField Directivity (Phi=90)



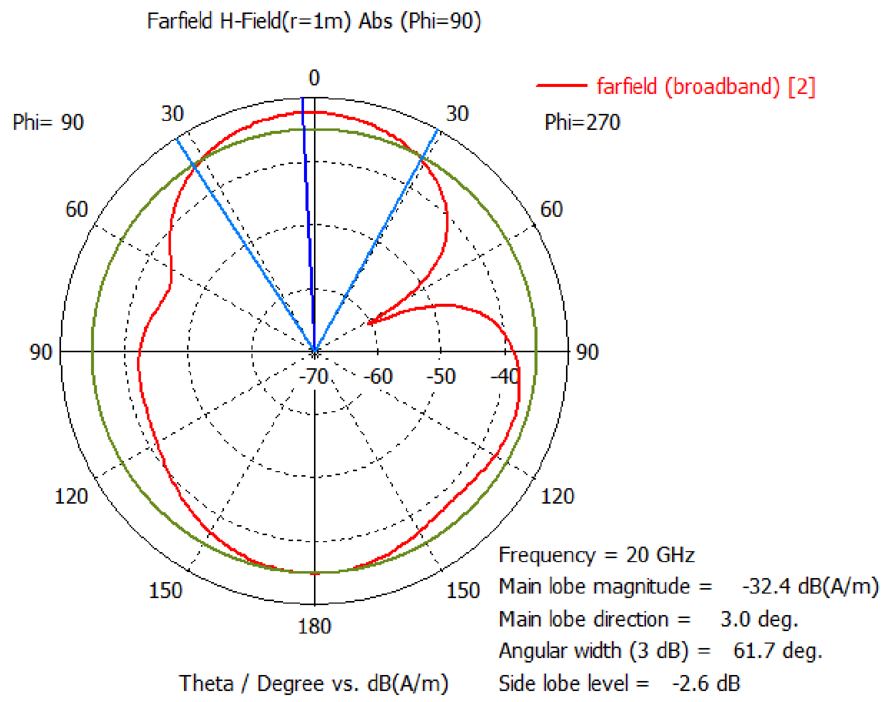
FarField E Field(Phi=0)



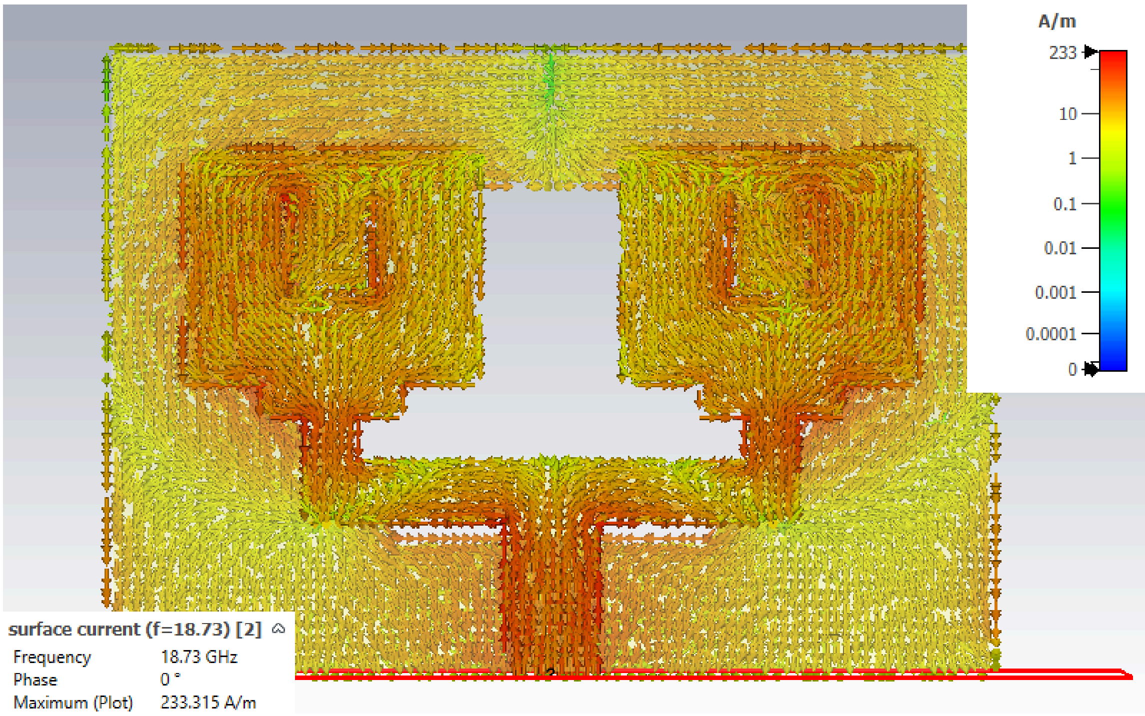
FarField E Field(Phi=90)



FarField H Field(Phi=0)



FarField H Field(Phi=90)



Surface Current

Chapter 5. Conclusion

The work carried out in this project successfully demonstrates the design, simulation, and validation of a **dual-band millimeter-wave microstrip antenna**, optimized to operate efficiently at **28 GHz** and **38 GHz**, the two most widely adopted bands for 5G NR FR2 applications. The study focused exclusively on the **single-element antenna** and its extension into a **1×2 array configuration**, ensuring that the design meets the fundamental requirements of mmWave systems such as compactness, high gain, stable radiation patterns, and strong impedance matching.

A key achievement of the antenna element is the use of **Rogers RT/Duroid 5880 substrate**, whose low permittivity and extremely low loss tangent significantly reduced dielectric losses at mmWave frequencies. This enabled clean dual-band resonance without distortion or frequency drift. Structural modifications—particularly the introduction of a **slot on the radiating patch** and **geometric tuning**—allowed the antenna to support two well-separated resonant modes using a single compact structure. The optimized feed-line taper further ensured excellent impedance matching, as evidenced by the deep **S11 dips** near both operating frequencies.

Simulation results confirm that the antenna maintains **broadside, stable radiation patterns** in each band. At 28 GHz, the antenna radiates with characteristics similar to the TM_{10} mode, whereas at 38 GHz the radiation becomes more directive due to higher-order current distribution. Correspondingly, the realized gain values of **6–7 dBi at 28 GHz** and **7–8 dBi at 38 GHz** indicate highly efficient radiation performance for a planar microstrip structure at mmWave frequencies.

Extending the design to a **1×2 antenna array** further enhances directivity and demonstrates the scalability of the proposed geometry. The array achieves significantly improved forward gain due to constructive interference between elements, while still maintaining strong impedance matching. The introduction of a **defected ground structure (DGS)** effectively reduces ground-surface currents, yielding noticeably lower coupling and cleaner radiation in the array configuration compared to a conventional structure.

Overall, the completed work—limited to the design and analysis of the single-element antenna and the 1×2 array—validates that compact, dual-band microstrip antennas can be engineered for reliable mmWave operation. The antenna offers strong dual-band performance, robust gain, stable patterns, and efficient fabrication feasibility, making it a practical solution for future 5G transceivers, small-cell systems, and directional mmWave modules. This foundation provides a solid platform for future advancements such as larger arrays, reconfigurable structures, or full array integration

Chapter 6. Future Works and Enhancements

Although the designed single-element and 1×2 array antenna successfully meets the essential performance requirements for 5G millimeter-wave bands at 28 GHz and 38 GHz, several opportunities remain for further improvement and expansion of the current work. These enhancements can significantly strengthen the antenna's applicability in future communication systems while maintaining the simplicity and practicality of the present design.

One key direction for future development is the incorporation of **reconfigurability** into the antenna structure. Technologies such as MEMS switches, PIN diodes, or varactor elements can enable dynamic tuning of frequency, bandwidth, or radiation characteristics. Integrating such mechanisms would allow the antenna to adapt to different mmWave channels, compensate for device orientation, or support multiple service bands using the same hardware—an increasingly valuable capability as 5G evolves toward more flexible and adaptive communication environments.

Another promising area is the **optimization of the array configuration**. While the current 1×2 array demonstrates improved gain and stable radiation patterns, exploring alternative array layouts such as 1×4 or 2×2 structures (without focusing on array behavior) can enhance directivity and provide more refined beam shaping. As array size increases, maintaining isolation and controlling surface-wave propagation will become critical, and these aspects can be addressed through improved ground-plane engineering, refined spacing strategies, or more advanced slot geometries.

Fabrication enhancements also offer significant room for progress. Although standard PCB processes were sufficient for this prototype, **advanced manufacturing methods**—including 3D printing, additive metal deposition, or inkjet-printed conductors—can enable more compact, lightweight, and flexible antenna structures. Exploring alternative substrates such as liquid crystal polymer (LCP), flexible laminates, or low-profile composite materials could further improve mechanical robustness and enable integration into wearable or conformal devices. Ensuring that these materials perform reliably at mmWave frequencies would form a valuable direction for extended research.

From a performance standpoint, additional electromagnetic characteristics may be explored, such as **circular polarization** or **polarization diversity**, which help reduce signal degradation caused by multipath reflections and orientation mismatches. Techniques like corner truncation, slotted perturbations, or dual-feed configurations can be studied to achieve these enhancements while maintaining compact geometry.

The defected ground structure (DGS) used in this work proved highly effective, but further experimentation with **advanced DGS geometries**—including fractal patterns, spiral cuts, or periodic frequency-selective shapes—may offer additional optimization in bandwidth, radiation purity, or array mutual coupling control. Incorporating metasurface-inspired structures above or below the patch may also provide opportunities to improve gain or tailor the radiation pattern without significantly increasing size.

On the simulation front, more advanced **multi-physics modeling** can be introduced. Since mmWave antennas are highly sensitive to mechanical and environmental variations, simulations that include thermal effects, substrate stress, or humidity variations can help predict real-world performance with greater accuracy. Machine-learning-driven optimization or surrogate models may additionally streamline the design cycle by rapidly identifying optimal geometrical parameters.

Finally, more extensive measurement and validation can be performed. Measurements under varying temperatures, different mounting substrates, and non-ideal environments would help assess stability and robustness. Over-the-air testing using mmWave communication testbeds can further validate practical performance in data transmission, link reliability, and real-world deployment scenarios.

In summary, while the present work successfully demonstrates a compact and efficient dual-band mmWave antenna and its 1×2 array extension, numerous opportunities exist to enhance, expand, and refine the design. Through improvements in reconfigurability, array scaling, material innovation, fabrication methods, and advanced modeling, the antenna can evolve into an even more versatile and powerful solution for future high-frequency wireless communication systems.

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